Bahria University Lahore Campus

Department of Computer Science

Course Code/number	**CSC 315**
Course Title/Name	**Theory of Automata**
Credit Hours/Contact hours	**3**
Degree Program	BSCS
Prerequisites or Co-requisites	Good programming skills, good analytical skills
Assessment Methods and Weightage	Quizzes 10 Assignments/Projects 20 Mid-Term Examination 20 Final Examination 50 **Total 100**
Textbook (or **Laboratory Manual** for Laboratory Courses)	Introduction To Computer Theory Daniel I.A Cohen 2nd Edition
Reference Material (With Edition, ISBN#)	Introduction to Automata Theory Languages and Computation by Hopcroft, Ulman
Web Resources/ URL (if any)	
Instructor Name/Cluster Head/Subject Expert	Instructor Name: Tahir Iqbal Designation: Snr. Assistant professor Status □ Regular Cluster Head Name: Dr. Iram Noreen
Course Aims	
Course Objectives	This course introduces students with the fundamental concepts of automata theory and formal languages, to form basic models of computation which provide foundation of many branches of computer science, e.g., compilers, software engineering, concurrent systems, etc.

Course Learning Outcomes	After successful completion of this course, the students should be able to:			**PLO**	**BT Level**
	1. **Explain** the different concepts in automata theory and formal languages. 2. **Analyze** properties of languages, grammars, and automata with formal methods. 3. **Design** grammars and models for different languages			2 3 4	C2 C4 C5
Course Description/Catalogue					
Lecture Plan (16 Weeks)	Week # and Date	Lecture/ Hour #	Topic to be covered	Learning outcomes	Reference Text
	1	1	Introduction to the course, Formal and In-formal languages,	Students will learn about the basics e.g. Alphabets, Strings, Null string, Words, Valid and In-valid alphabets, length of a string, Reverse of a string,	Chapter 1 and Lecture Handouts
		2	Defining languages, Descriptive definition of languages	Students will learn different ways to define language	Chapter 2 and Lecture Handouts
	2	1	Kleene Star Closure, Plus operation	Students will concept of Kleene start and Plus in Languages	Chapter 3 and Lecture Handouts
		2	Recursive definition of languages	Students can give recursive definition of languages, INTEGER, EVEN, factorial, PALINDROME	Chapter 3 and Lecture Handouts
	3	1	Recursive definition of RE, defining languages by Regular expression	An understanding of defining languages by RE e.g. **exactly one aa, even**	Chapter 3 and Lecture Handouts

				length, **odd length**, **at least one a**	
		2	Regular expression language, Difference between a* + b* and (a+b)*,	An understanding of Regular expression of **EVEN-EVEN**	Chapter 4 and Lecture Handouts
	4	1	Equivalent regular expressions;	Student will learn Equivalent regular expressions	Chapter 4 and Lecture Handouts
		2	sum, product and closure of regular expressions regular languages,	Student will understand Different operations on RE's	Chapter 4 and Lecture Handouts
	5	1	introduction to finite automaton, definition of FA, transition table, transition diagram	Student will understand Finite automata and its basics	Chapter 5 and Lecture Handouts
		2	Different notations of transition diagrams, FA corresponding to finite languages	An understanding of FA corresponding to RL	Chapter 5 and Lecture Handouts
	6	1	Transition graphs **definition, TG Examples**	Understanding of TG Sharpen hands on **TG Examples**	Chapter 6 and Lecture Handouts
		2	Generalized Transition Graphs	Learn Basics of GTG's	Chapter 6 and Lecture Handouts
	7	1	Proof of Kleene's theorem part I	A good understanding of Process Synchronization	Chapter 7 and Lecture Handouts
		2	Proof of Kleene's theorem part II (method with different steps), Particular examples of TGs to determine corresponding REs.	Learn examples of TGs , TG's to determine corresponding REs.	Chapter 7 and Lecture Handouts

	8	1	Kleene's theorem part III (method 1:union of FAs)	An understanding of Union of FA's	Chapter 7 and Lecture Handouts
		2	method 2 : Concatenation of FAs method 3 : closure of an FA	Kleene's theorem with different methods	Chapter 7 and Lecture Handouts
			M I D T E R M		
	9	1	Nondeterministic Finite Automaton (NFA)	Basic Understanding of NDFA	Chapter 8 and Lecture Handouts
		2	Converting an FA to an equivalent NFA method (1,2)	Learn FA to NFA different conversion methods	Chapter 8 and Lecture Handouts
	10	1	Converting an FA to an equivalent NFA method 3	Explore other methods of NFA to FA conversion	Chapter 8 and Lecture Handouts
		2	Moore machine	An understanding of Finite Automaton with output,	Chapter 9 and Lecture Handouts
	11	1	Mealy machine	Basic understanding of Mealy machine	Chapter 9 and Lecture Handouts
		2	Complementing machine, Incrementing machine.	Student will understand Complementing and incrementing machines and their Applications	Chapter 9 and Lecture Handouts
	12	1	Equivalent machines , Moore equivalent to Mealy, Mealy equivalent to Moore	Students will learn proof and examples of machine conversions	Chapter 9 and Lecture Handouts
		2	Mealy machines in terms of sequential circuit	Understanding of mealy as a sequential circuit	Chapter 9 and Lecture Handouts
	13	1	Regular languages, complement of a language, intersection of two regular languages	An introduction to complement of Regular languages, theorem and proof	Chapter 10 and Lecture Handouts

		2	non regular language, pumping lemma version I	An introduction to non-regular language, pumping lemma proof and examples	Chapter 11 and Lecture Handouts
	14	1	Pumping lemma version II Myhill Nerode theorem	Students will understand Pumping lemma version II, proof and examples	Chapter 11 and Lecture Handouts
		2	Decidablity	Students will learn about equivalence of FA's	Chapter 12 and Lecture Handouts
	15	1	Turing Machine	Students will Learn Turing Machine concept	Chapter 24 and Lecture Handouts
		2	Context Free Grammar, Terminals, non-terminals, productions, CFG, context Free language, Parse Tree	Example of trees, Polish Notation, examples, Ambiguous CFG, example,	Chapter 13 and Lecture Handouts
	16	1	Pushdown Automata	Understanding of Pushdown Automata	Chapter 14 and Lecture Handouts
		2	Equivalence of PDA and CFG Deterministic Vs non-Deterministic PDA	An comparative analysis of Deterministic Vs non-Deterministic PDA	Chapter 16 and Lecture Handouts
			Final Exam		

Semester Calendar for Assignments/Quizzes/Project

Assignments/ Projects and quizzes Plan	Week #	Assignment No.	Quiz No.	Project Title	Assignment/Project Quiz Date	Result Date of Assignment/Project/Quiz
	1					
	2					
	3		Quiz			Quiz
	4	Assignment-1				
	5				Assignment-1	
	6		Quiz			Assignment-1
	7	Assignment-2			Assignment-2	Quiz
	8		Quiz			Assignment-2 Quiz
	10	Assignment-3				
	11		Quiz		Assignment-3	
	12					
	13					Assignment-3, Quiz,
	14	Assignment-4	Quiz			Quiz

PLOs/CLOs	CLO1	CLO2	CLO3
PLO1 (Academic Education)			
PLO2 (Knowledge for Solving Computing Problems)	✓		
PLO3 (Problem Analysis)		✓	
PLO4 (Design/Development of Solutions)			✓
PLO5 (Modern Tool Usage)			
PLO6 (Individual and Teamwork)			
PLO7 (Communication)			
PLO8 (Computing Professionalism and Society)			
PLO9 (Ethics)			
PLO10 (Life-long Learning)			

Instructor Name __TAHIR IQBAL_____

Instructor Signature _____

Date _____